Adaptive Sub-Stepping Evaluation — Correction Report

1. Purpose

This document reviews the before/after adaptive sub-stepping experiment and proposes targeted corrections to improve technical clarity, consistency, and reviewer robustness before final submission.

2. Summary of Findings

The adaptive sub-stepping experiment is well-designed and isolates a single variable: enabling vs disabling adaptive sub-stepping under identical simulation conditions. The results clearly demonstrate that adaptive sub-stepping eliminates timestep-induced instability and significantly improves accuracy.

Key validated outcomes:

• Worst-case error reduction: 161.7 AU → 1.543 AU (~105× improvement)

• Divergence rate reduction: ~0.251 → ~0.156 /yr (~38% reduction)

• Stability improvement: Ejection → Fully bounded system

• Computational cost: +35% per-step overhead

• Activation frequency: only ~0.2–1.1% of steps

3. Identified Issues

Issue 1: Divergence rate inconsistency

Different parts of the project report slightly different post-adaptive divergence rates (~0.156 vs ~0.1655 /yr).

Issue 2: Interpretation wording

The report currently describes the improvement as a 'physical improvement', which is not technically precise.

4. Proposed Corrections

Correction 1: Clarify divergence rate reporting

Suggested wording:

The divergence rate after adaptive sub-stepping lies in the range 0.156–0.166 /yr depending on the fit window and summary run. The qualitative conclusion remains unchanged.

Correction 2: Replace interpretation language

Replace: genuine physical improvement

With: numerical stability improvement due to improved resolution of close encounters.

5. Strengthened Final Claim (Recommended)

Adaptive sub-stepping is the dominant stability mechanism in SIMON. It eliminates timestep-induced ejection during close encounters, reduces worst-case trajectory error by over two orders of magnitude, and lowers the divergence rate, while introducing only modest computational overhead and activating in a small fraction of simulation steps.

6. Final Recommendation

The experiment is strong and publication-ready after incorporating the above corrections. The results provide clear, quantitative evidence for the role of adaptive sub-stepping and significantly strengthen the paper’s engineering contribution.